IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

**ANALYSIS AND DESIGN OF CIRCUITS**

Thursday, 1 May 2025, 2:30 PM

Time allowed: 2:00 hours

**There are THREE questions on this paper.**

**Answer ALL questions.**

*Question One carries 40% of the marks. Question 2 and 3 carry equal marks (30% each).*

**Any special instructions for invigilators and information for candidates are on page 1.**

Examiners responsible First Marker(s) : Z. Akhtar, P. Mitcheson
Second Marker(s) :

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# Information for Candidates

## Numerical Answers

- Numerical answers should be given as fully evaluated decimal values and not as unevaluated arithmetic expressions.

## Notation

The following notation is used in this paper:

- The voltage waveform at node $X$ in an AC circuit is denoted by $x(t)$, while the phasor voltage is denoted by $X$.
- Component and source values are generally given in Ohms, Farads, Henrys, Volts, or Amps, unless otherwise stated.
- Times are in seconds, unless otherwise stated.

1. (a) Using nodal analysis, determine the voltages at nodes $X$ and $Y$ in the circuit shown in Figure 1a. Then, calculate the Thévenin equivalent circuit for the entire network (including the 6 kΩ resistor) between nodes $X$ and $Y$. [6]

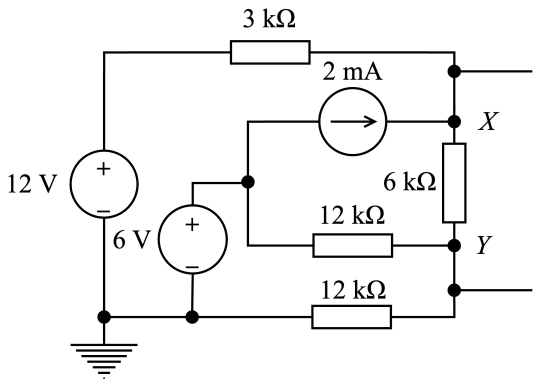



Figure 1a

(b) For the filter circuit shown in Figure 1b, determine the transfer function $H(j\omega) = \dfrac{Y(j\omega)}{X(j\omega)}$. [8]

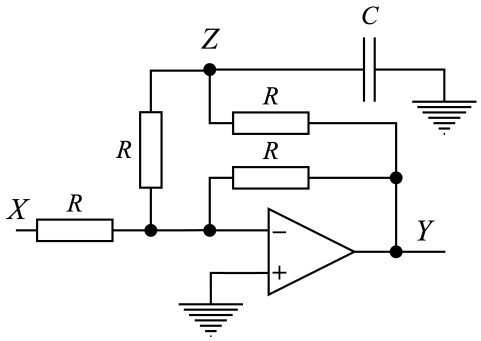



Figure 1b

(c) For the circuit shown in Figure 1c, assume the diode has a forward voltage drop of 0.7 V. Determine the value of $U$ at which the diode switches between operating regions. Write an expression for the output voltage $Y$ in terms of $U$ for both operating modes. [6]

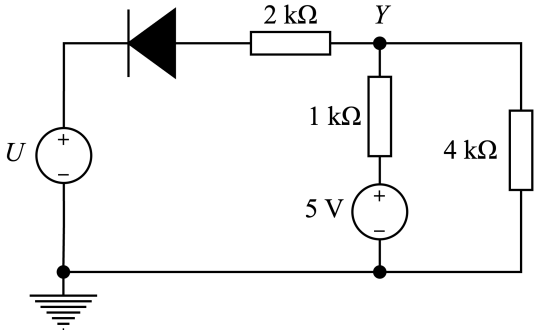



Figure 1c

(d) State the ideal properties of an amplifier and the limitations in achieving these with a single-stage common emitter configuration. [8]

(e) Calculate the small signal voltage gain of the amplifier shown in Figure 1e. You may assume that the the capacitor $C_E$ is a short circuit at signal frequencies. [7]

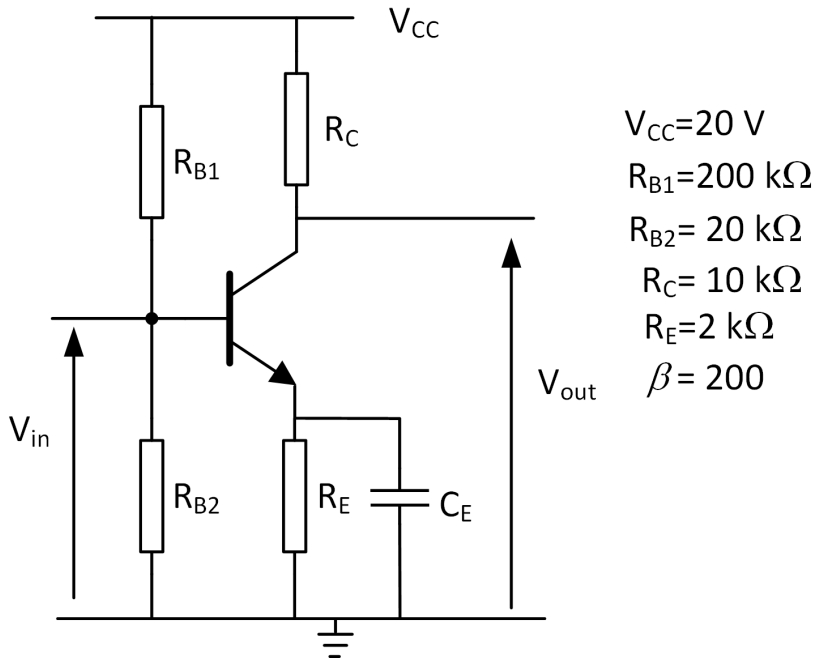



Figure 1e

(f) Calculate the quiescent output voltage in the circuit of Figure 1f. [5]

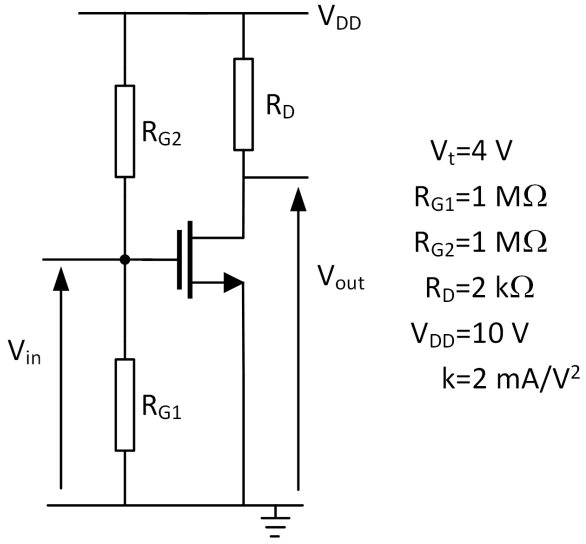



Figure 1f

2. For the circuit shown in Figure 2, answer the following:

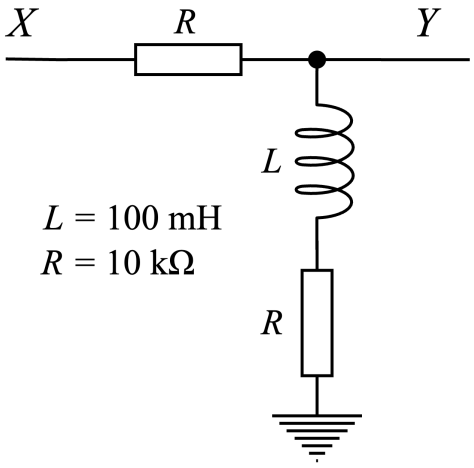



Figure 2

(a) Find the transfer function $H(j\omega) = \dfrac{Y(j\omega)}{X(j\omega)}$. [6]

(b) Sketch a straight-line (asymptotic) approximation of the frequency response between 100 Hz and 1 MHz. Both the magnitude and phase plots are required. Each plot carries half of the total marks. [8]

(c) Given that the input phasor is $X = 5 - 5j$ , and assuming an angular frequency of $\omega = 2\pi \times 10^3$ rad/s, determine the output phasor $Y$. [6]

(d) Obtain an expression for $y(t)$ when the input signal $x(t)$ is as shown in Figure 2d, with: [10]

$$x(t) = \begin{cases} 5\sin(400000\pi t), & t < 0 \\ 10\cos(200000\pi t), & t \geq 0 \end{cases}$$

The expression for $y(t)$ should include the steady-state response before and after the switching at $t = 0$, as well as any transient response generated by the switching.

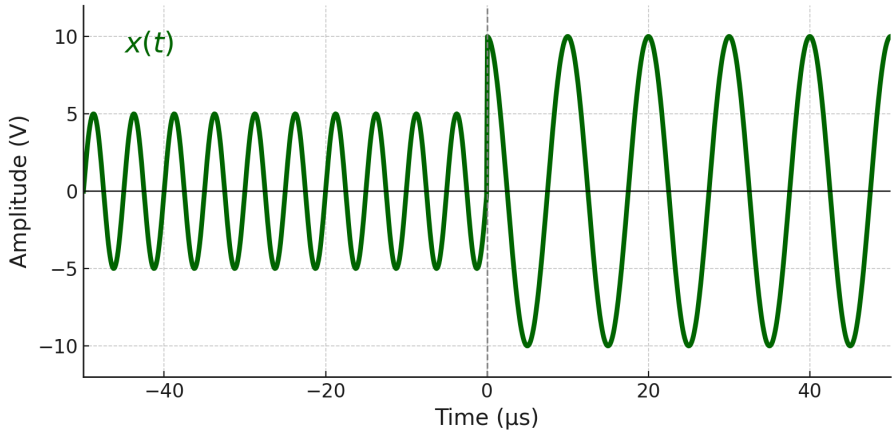



Figure 2d

3. This question concerns the differential amplifier show in Figure 3. All transistors are matched.

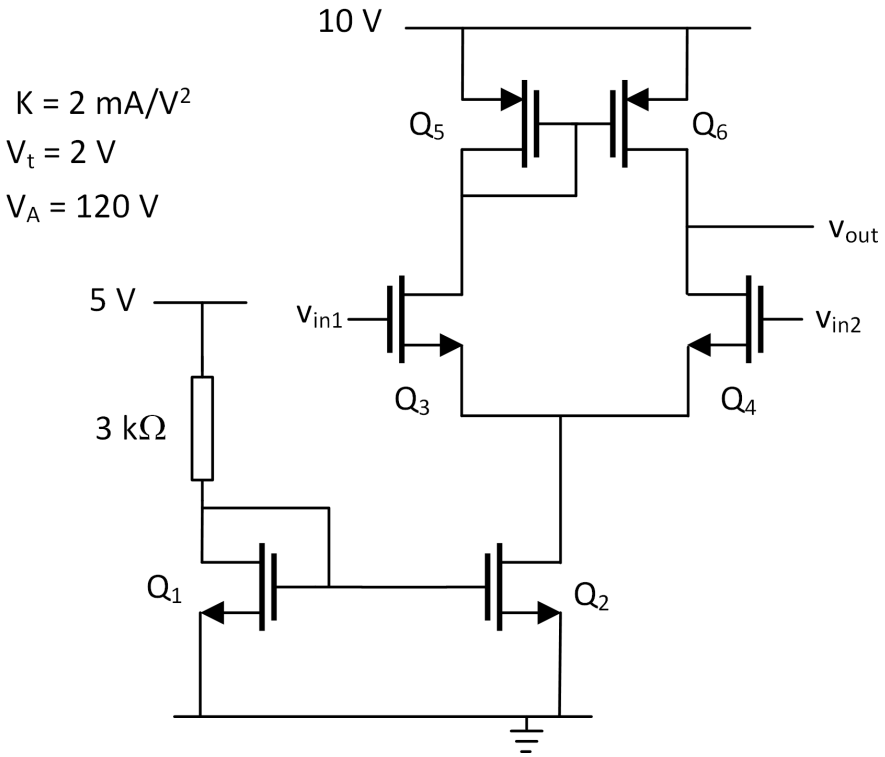



Figure 3

(a) Define and explain the terms differential-mode gain, common-mode gain, and common mode rejection ratio (CMRR). [7]

(b) Calculate the tail current (the drain current of $Q_2$) in the circuit of Figure 3. Ignore channel length modulation in your calculation. [5]

(c) Draw a small signal model of the differential amplifier under differential mode excitation (i.e. when $v_{in1} = -v_{in2} = v_d/2$). [4]

(d) Calculate the small signal voltage gain and state any assumptions. [5]

(e) Calculate the minimum common-mode voltage on the inputs so that the amplifier remains within its common mode operating range. [3]

(f) It is proposed to add a BJT based class AB stage to the output of the differential amplifier. Draw a class AB stage and explain: [6]

- How cross-over distortion is almost eliminated.
- How shoot-though current between the power supply rail and ground is controlled.